1-to'plam. *Daraja qoidalari – nol/manfiy ko'rsatkichsiz sof mashq.*

1. $2^2 \times 2^3 =$

2. $\frac{3^4}{3^2} =$

3. $(2^2)^3 =$

4. $5^3 \times 5^2 =$

5. $\frac{4^5}{4^3} =$

6. $(3^2)^2 =$

7. $2^3 \times 2^3 =$

8. $(2^3)^3 =$

2-to'plam. *Nol va manfiy ko'rsatkichlar.*

1. $5^0 =$

2. $2^{-3} =$

3. $4^{-2} =$

4. $2^0 \times 2^{-2} =$

5. $3^{-2} \times 3^4 =$

6. $5^2 \times 5^{-2} =$

7. $10^{-1} \times 100 =$

8. $2^{-1} + 2^{-2} =$

3-to'plam. *Aralash – haqiqiy test zichligi.*

1. $\dfrac{(2^2 \times 2^3)^2}{2^4} =$

2. $(3 \times 3^2)^2 =$

3. $\dfrac{5^3}{5^2 \times 5^{-1}} =$

4. $(2^{-1})^{-3} =$

5. $(4^2)^0 =$

6. $\dfrac{2^5}{2^{-2}} =$

7. $\dfrac{(3^2)^2}{3^3} =$

8. $10^3 \times 10^{-1} \div 10 =$

4-to'plam. *Kvadrat ildizlarni soddalashtirish.*

1. $\sqrt{16} =$

2. $\sqrt{49} =$

3. $\sqrt{18} =$

4. $\sqrt{50} =$

5. $\sqrt{72} =$

6. $\sqrt{20} =$

7. $\sqrt{45} =$

8. $\sqrt{8} =$

5-to'plam. *Ildizlarni qo'shish va ayirish.*

1. $\sqrt{8} + \sqrt{2} =$

2. $\sqrt{18} - \sqrt{2} =$

3. $\sqrt{12} + \sqrt{27} =$

4. $\sqrt{50} - \sqrt{8} =$

5. $2\sqrt{3} \times \sqrt{3} =$

6. $\sqrt{2} \times \sqrt{8} =$

7. $3\sqrt{5} + 2\sqrt{5} =$

8. $\sqrt{45} - \sqrt{20} =$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $2^3 \times 2^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$2^3 \times 2^3 = 2^9 = 512$."
   Xato va to'g'ri javob: .....

2. $(2^2)^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$(2^2)^3 = 2^{2+3} = 2^5 = 32$."
   Xato va to'g'ri javob: .....

3. 2^{-3} ifodasida ko'rsatilgan yechimda xato bor:
   "$2^{-3} = -8$."
   Xato va to'g'ri javob: .....

4. $\sqrt{16+9}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$."
   Xato va to'g'ri javob: .....

5. $\sqrt{32}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$32 = 16 \times 2$ *deb to'g'ri ajratilgan, lekin javobda* $16\sqrt{2}$ *deb yozilgan.*"
   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\dfrac{(2^3)^2}{2^4} + \sqrt{16} =$

2. $\sqrt{3^2 \times 2^2} =$

3. $2^{-2} \times 4^2 + \sqrt{9} =$

4. $(\sqrt{2})^4 =$

5. $1 + \sqrt{5^2 - 3^2} =$

8-to'plam. *SAT'da qanday uchraydi – daraja va ildiz qoidalarini qo'llash.*

1 $2^3 \times 2^4$ ifodaning qiymati nechaga teng?

(A) 128 (B) 64

(C) 16 (D) 81

2 $\frac{5^6}{5^2}$ ifodaning qiymati nechaga teng?

(A) 25 (B) 625

(C) 125 (D) 15625

3 $(x^3)^4$ ifodasi barcha x uchun qaysi ifodaga teng?

(A) x^7 (B) x^{64}

(C) x^{12} (D) x^{81}

4 $7^0 + 3^2$ ifodaning qiymati nechaga teng?

(A) 9 (B) 16

(C) 19 (D) 10

5 4^{-2} ifodaning qiymati nechaga teng?

(A) $\frac{1}{16}$ (B) -8

(C) -16 (D) 16

6 Agar $a = 2^5$ va $b = 2^3$ bo'lsa, $\frac{a}{b}$ nechaga teng?

(A) 2 (B) 4

(C) 8 (D) 16

7 $\sqrt{98}$ ni eng oddiy ildiz ko'rinishida yozing.

(A) $14\sqrt{2}$ (B) $2\sqrt{49}$

(C) $7\sqrt{2}$ (D) $49\sqrt{2}$

8 Kvadratning yuzi 48 kvadrat birlikka teng. Uning tomoni (eng oddiy ildiz ko'rinishida) nechaga teng?

(A) $16\sqrt{3}$ (B) $6\sqrt{2}$

(C) $2\sqrt{12}$ (D) $4\sqrt{3}$

9 $3^4 \times 3^{-2}$ ifodaning qiymati nechaga teng?

(A) 9 (B) 729

(C) $\frac{1}{9}$ (D) 1

10 $2^n = \frac{1}{8}$ tenglamasini qanoatlantiruvchi n qanday qiymatga ega?

(A) 3 (B) -3

(C) -8 (D) 8

11 $\sqrt{45} + \sqrt{20}$ ifodaning qiymati nechaga teng?

(A) $5\sqrt{65}$ (B) $\sqrt{65}$

(C) $5\sqrt{5}$ (D) $25\sqrt{5}$

12 Idishdagi bakteriyalar soni har soatda ikki baravar oshadi. Boshlang'ich son $P_0 = 2^3$. Yana 2 marta ikki baravar oshgandan keyin, soni (daraja ko'rinishida) nechaga teng?

(A) 2^6 (B) 2^1

(C) 2^9 (D) 2^5

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	32	2	9
	3	64	4	3125
	5	16	6	81
	7	64	8	512
2-to'plam	1	1	2	$\frac{1}{8}$
	3	$\frac{1}{16}$	4	$\frac{1}{4}$
	5	9	6	1
	7	10	8	$\frac{3}{4}$
3-to'plam	1	64	2	729
	3	25	4	8
	5	1	6	128
	7	3	8	10
4-to'plam	1	4	2	7
	3	$3\sqrt{2}$	4	$5\sqrt{2}$
	5	$6\sqrt{2}$	6	$2\sqrt{5}$
	7	$3\sqrt{5}$	8	$2\sqrt{2}$
5-to'plam	1	$3\sqrt{2}$	2	$2\sqrt{2}$
	3	$5\sqrt{3}$	4	$3\sqrt{2}$
	5	6	6	4
	7	$5\sqrt{5}$	8	$\sqrt{5}$
7-to'plam	1	8	2	6
	3	7	4	4
	5	5	–	–
8-to'plam	1	A	2	B
	3	C	4	D
	5	A	6	B
	7	C	8	D
	9	A	10	B
	11	C	12	D
6-to'plam	1. Ko'rsatkichlarni ko'paytirgan, qo'shish kerak edi (ko'paytma qoidasi). To'g'ri: $2^{3+3} = 2^6 = 64$.			
	2. Ko'rsatkichlarni qo'shgan, ko'paytirish kerak edi (daraja darajasi). To'g'ri: $2^{2\times 3} = 2^6 = 64$.			
	3. Manfiy ko'rsatkichni asosni manfiy qiladi deb tushungan. To'g'ri: $2^{-3} = \frac{1}{8}$.			
	4. Yig'indining ildizini ildizlar yig'indisiga bo'lgan – bu hech qachon to'g'ri emas. To'g'ri: $\sqrt{25} = 5$.			
	5. 16 ni to'g'ri ajratgan, lekin uning ildizini olmagan. To'g'ri: $\sqrt{16 \times 2} = 4\sqrt{2}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram